

BUSINESS INSIGHTS REPORT (EDA)

Task 04: Exploratory Data Analysis on Marketing Campaign Dataset

Objective

The objective of this project was to perform Exploratory Data Analysis (EDA) on a marketing campaign dataset to understand customer demographics, purchasing behavior, and campaign effectiveness. The analysis aims to provide actionable business insights and recommendations to improve marketing performance and return on investment (ROI).

Dataset Overview

The dataset contains customer information including age, income, marital status, product purchases, campaign responses, and spending patterns. Data cleaning was performed before analysis, including handling missing values and converting date columns into appropriate formats.

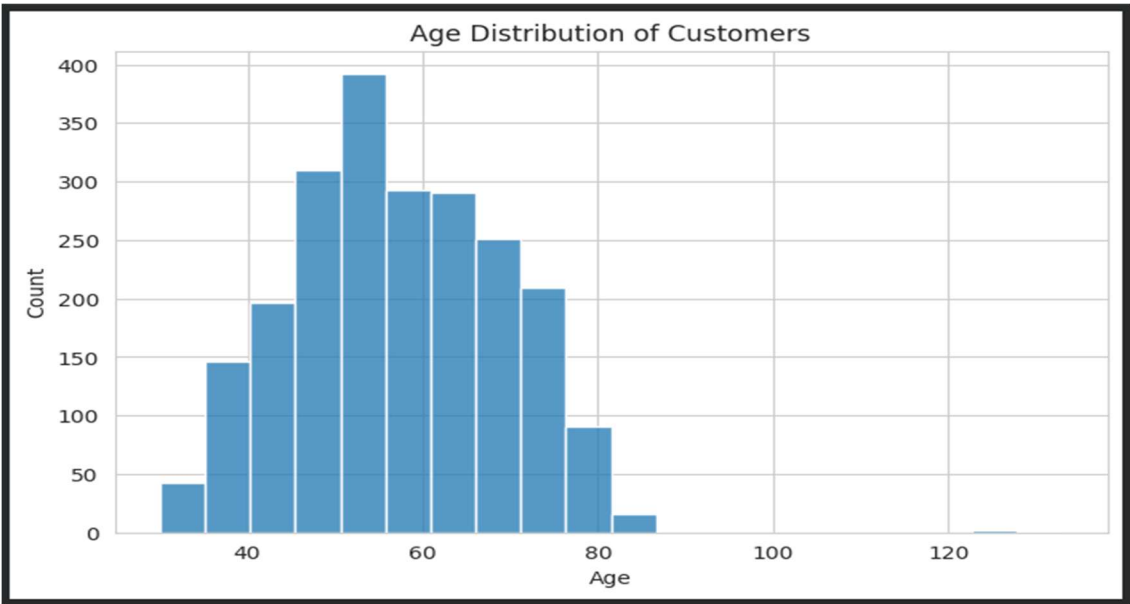
Analysis and Findings

1. Age Distribution of Customers

The age distribution analysis shows that most customers fall between 45 and 70 years of age, with the highest concentration around 50–60 years.

Insight:

The company’s primary customer base consists of middle-aged and older adults. Marketing campaigns should be tailored to the preferences and needs of this age group.

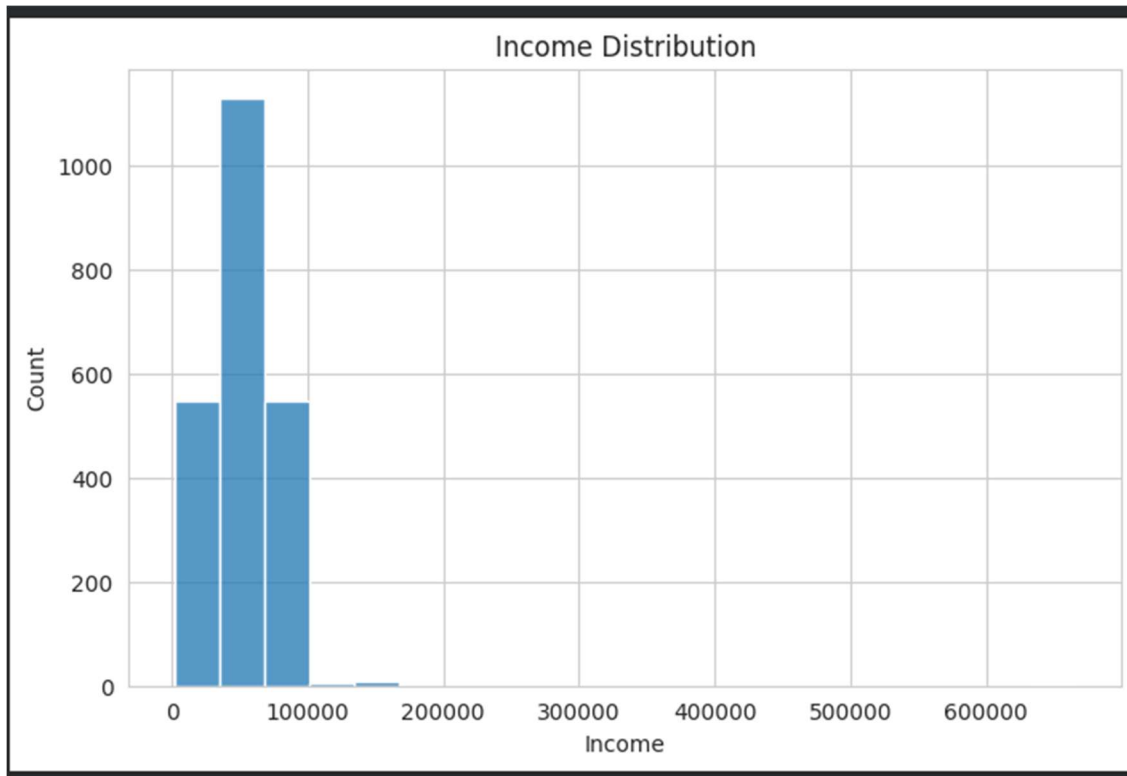


2. Income Distribution

The income distribution analysis indicates that most customers belong to the middle-income category, while a small number of customers have significantly higher incomes.

Insight:

The majority of customers have moderate purchasing power, but high-income customers represent a valuable segment for premium products and targeted marketing.



3. Campaign Response Analysis

The campaign acceptance analysis revealed the following performance:

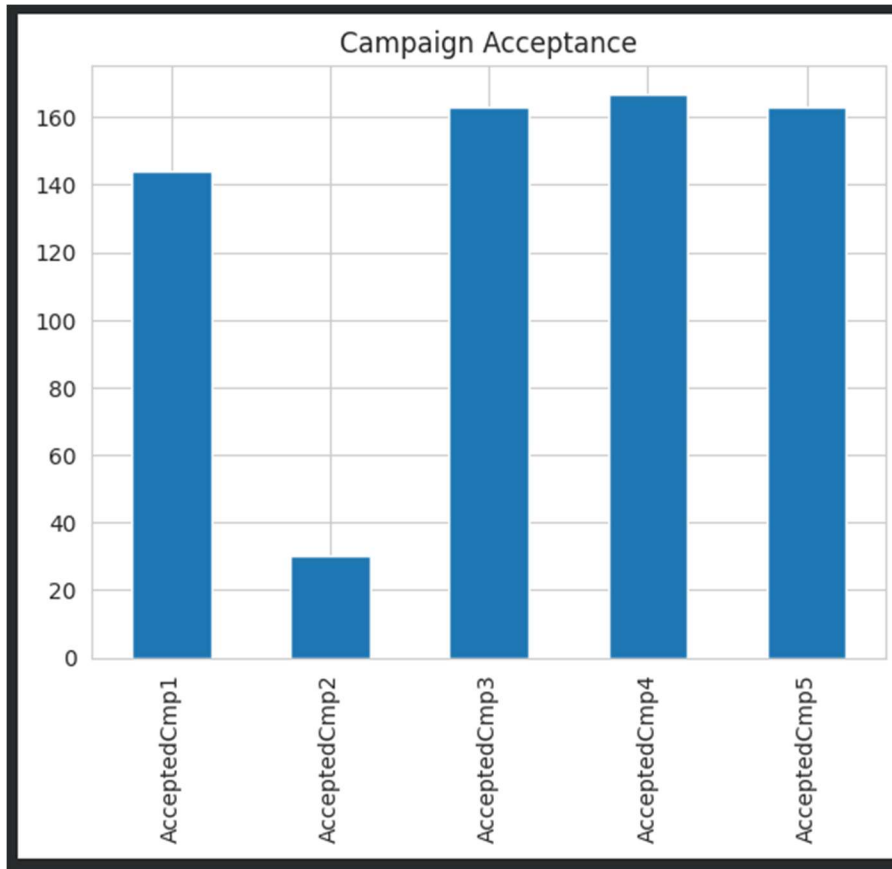
- Campaign 4 achieved the highest acceptance rate.
- Campaigns 3 and 5 also performed strongly.
- Campaign 2 recorded the lowest acceptance rate.

Insight:

Campaigns 3, 4, and 5 were more effective in attracting customer responses and engagement.

Recommendation:

Increase budget allocation for Campaigns 3, 4, and 5 while reviewing and redesigning Campaign 2 to improve performance.



4. Product Spending Analysis

Customer spending was highest in the following categories:

1. Wines
2. Meat Products
3. Gold Products

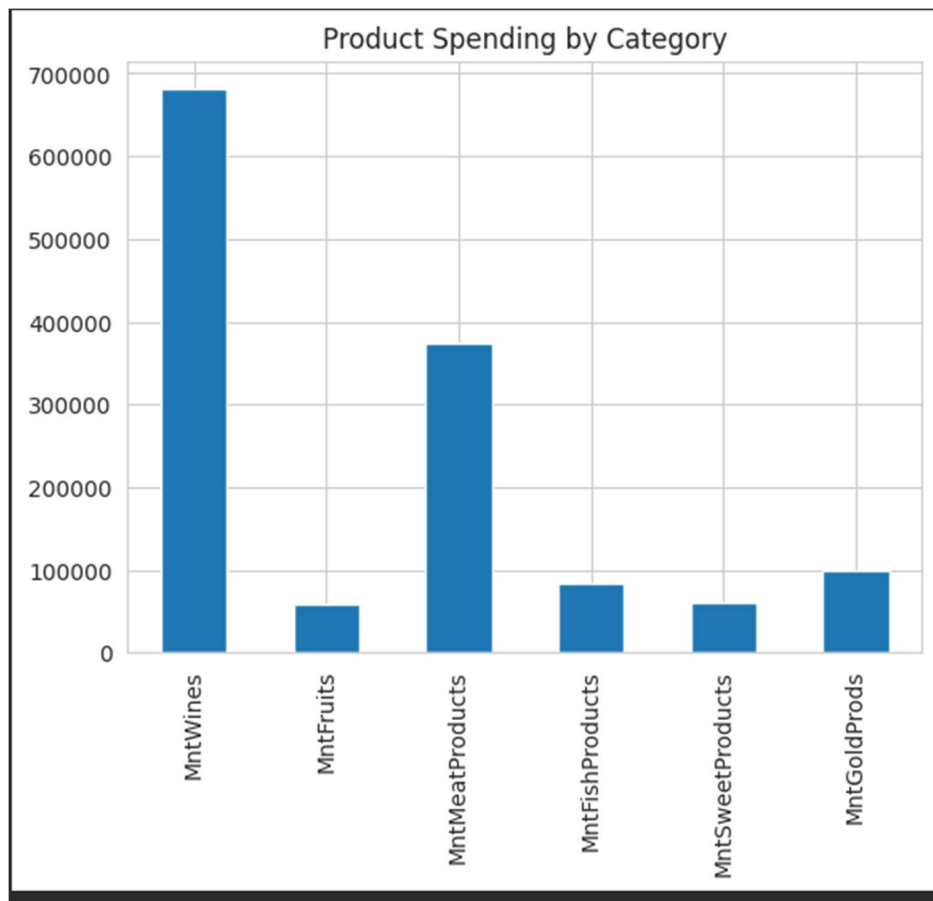
Lower spending was observed in Fruits, Fish Products, and Sweet Products.

Insight:

Wine and Meat Products generate the highest revenue contribution and customer interest.

Recommendation:

Focus future promotions and marketing efforts on Wine and Meat product categories to maximize sales and customer engagement.



5. Income vs Total Spending Analysis

The scatter plot analysis shows a positive relationship between customer income and total spending.

Average Values:

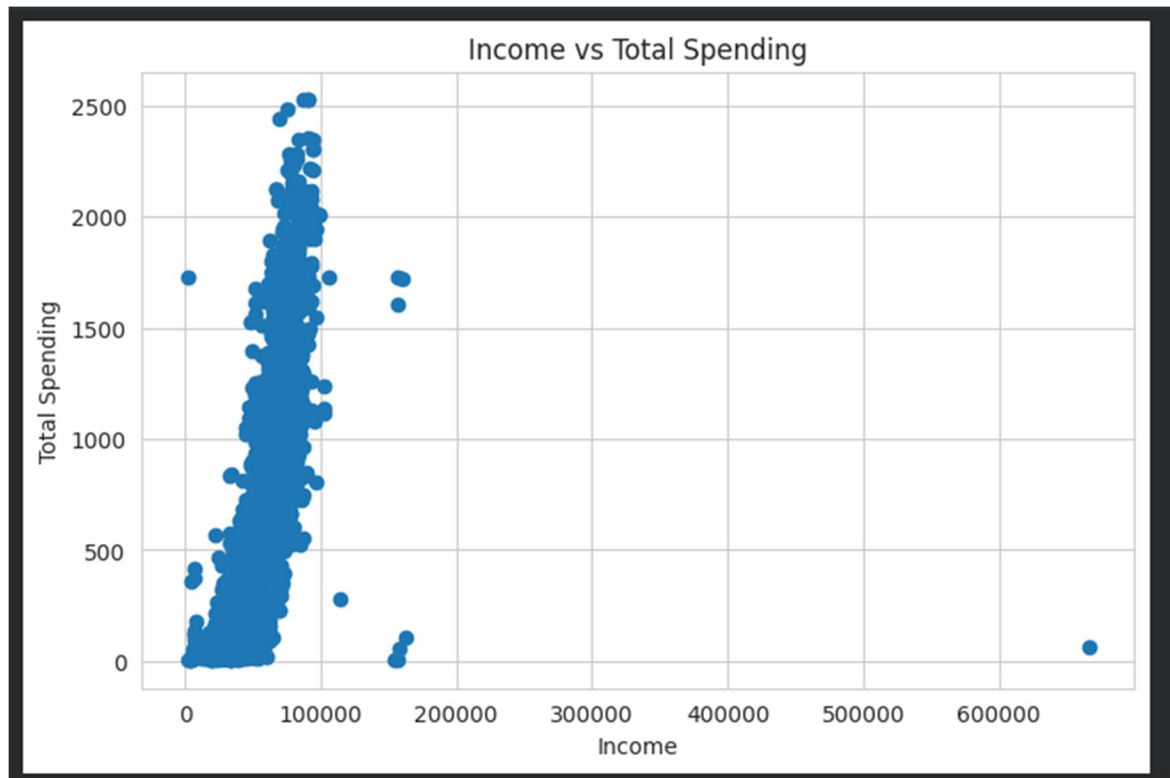
Response	Average Income	Average Spending
No Response (0)	50,846	539
Response (1)	60,183	987

Insight:

Customers who responded to marketing campaigns generally have higher incomes and spend significantly more than non-responders.

Recommendation:

Target high-income customers through personalized campaigns, premium product offers, and loyalty programs to increase revenue.



Business Recommendations

Based on the analysis, the following recommendations are proposed:

1. Increase marketing budget for Campaigns 3, 4, and 5 due to their higher acceptance rates.
 2. Review and redesign Campaign 2 to improve customer engagement.
 3. Focus promotional activities on Wine and Meat Products, which generate the highest customer spending.
 4. Develop targeted marketing strategies for high-income customers.
 5. Implement loyalty and retention programs for valuable customers to increase long-term profitability.
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Conclusion

The exploratory data analysis provided valuable insights into customer demographics, purchasing behavior, and campaign effectiveness. The results indicate that Campaigns 3, 4, and 5 are the most successful marketing initiatives, while Wine and Meat Products contribute the most to overall revenue. Additionally, higher-income customers exhibit greater spending behavior and represent a key target segment for future marketing efforts.

By focusing on high-performing campaigns, profitable product categories, and high-value customers, the company can improve marketing efficiency, increase ROI, and drive sustainable business growth.